Q3) A mobile OS has the following Tasks.

TASK	AT	BT
Music Player	0	20
Touch event	3	1
Whatsapp Notification	5	2
Camera launch	6	4

↳ Using this data answer the following.

1) which scheduling algorithm's most suitable.

2) Draw the GT for SRTF.

3) why Selected Algorithm from (1) improves user experience.

4) which task receives the fastest response.

Sol

1) SRTF – Shortest Remaining Time First

Because this is a mobile OS, short interactive tasks such as:

- touch events
- Notifications
- camera launch

⇒ Should receive CPU quickly rather than waiting behind a long task like the music player.

2) Ready Queue (RQ)

0:-

Music Player	

→ only Process available → selected

3:-

Music Player	touch event

→ MP remaining = 17
TE = 1
$1 < 17$ → Touch selected

4:-

~~Touch Event~~	Music Player

→ only remaining Process

5:-

Music Player	whatsapp Notification

→ MP remaining = 16
whatsapp = 2
$2 < 16$ → whatsapp selected

6:-

whatsapp Notification	

→ Whatsapp remaining = 1
camera = 4, Music = 16
$1 < 4 < 16$ → whatsapp continues

7:-

~~whatsapp Notification~~	Camera Launch

→ Camera = 4 < Music = 16

11:-

~~Camera Launch~~	Music player

→ only Process remaining

27:- DONE

Gantt chart (GT)

MUSIC	Touch	Music	Whatsapp	Camera

0 3 4 5 7 11 27

3)

↳ SRTF gives CPU quickly to short and interactive tasks, reducing their response time and making the mobile OS more responsive.

Short tasks → Quick Response → Better user experience

4) RT = First CPU Start Time − AT

Task	AT	First Start	RT
Music Player	0	0	0
Touch event	3	3	0
whatsapp Notification	5	5	0
camera Launch	6	7	1

⇒ Music Player, Touch event & whatsapp notification receive the fastest response.